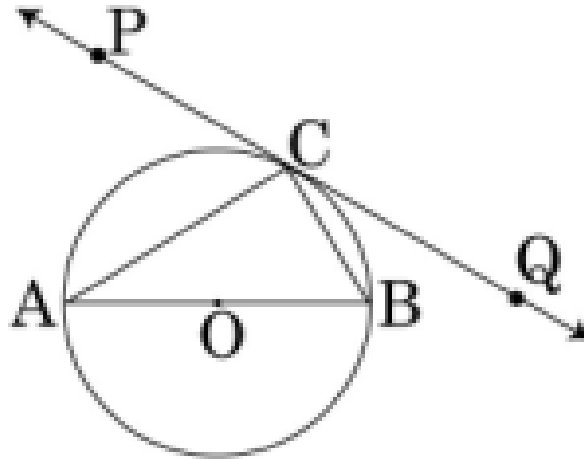


MATHEMATICS

SECTION – A

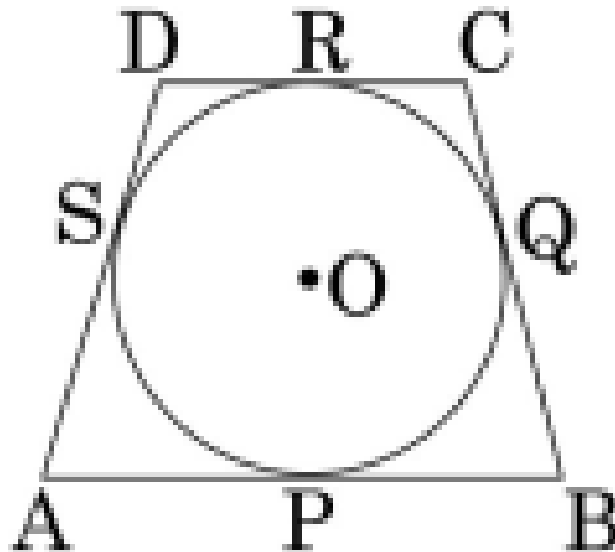
1. In Fig., PQ is a tangent at point C to a circle.



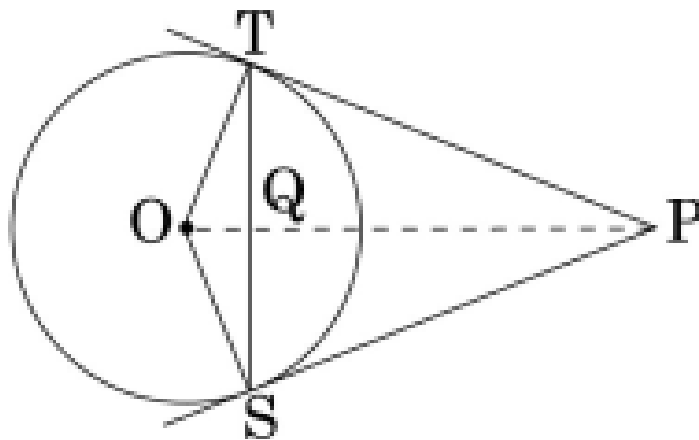
2. For what value of k will $k + 9$, $2k - 1$ and $2k + 7$ be consecutive terms of an A.P.?
3. A ladder leaning against a wall makes an angle with the horizontal. Find its length.
4. A card is drawn at random from a well-shuffled pack of cards. Find the probability of the required event.

SECTION – B

5. If -5 is a root of the quadratic equation $2x^2 + px - 15 = 0$ and the quadratic equation $p(x^2 + x) + k = 0$ has equal roots, find k .
6. Let P and Q be the points of trisection of the line segment joining $A(2, -2)$ and $B(-7, 4)$ such that P is nearer to A . Find coordinates of P and Q .
7. In Fig., a quadrilateral $ABCD$ circumscribes a circle. Prove that $AB + CD = BC + DA$.

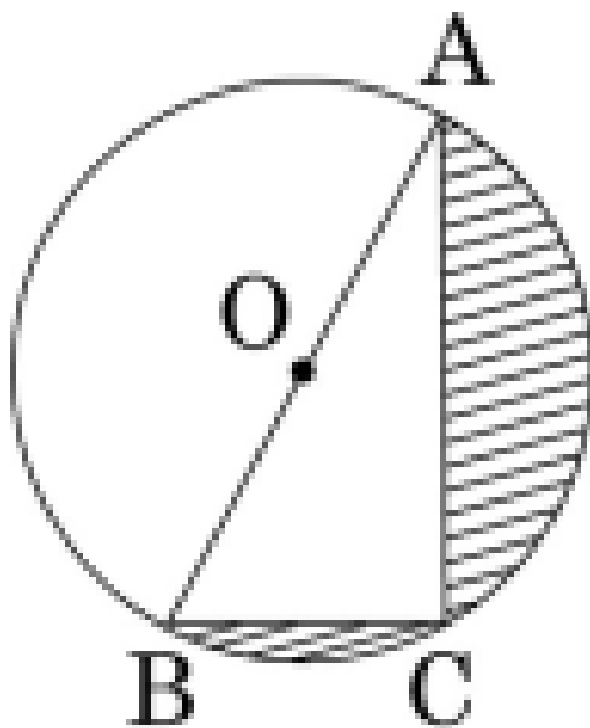


8. Prove that the points $(3, 0)$, $(6, 4)$ and $(-1, 3)$ form a right angled isosceles triangle.
9. The 4th term of an A.P. is zero. Prove that the 25th term is three times the 11th term.
10. From an external point P , two tangents are drawn to a circle. Show that $\angle OTS = \angle OST = 30^\circ$.



SECTION – C

11. In Fig., O is the centre of a circle such that required measurements are given. Find the area. Take $\pi = 3.14$.



- 12.** A tent is in the shape of a cylinder surmounted by a cone. Find the canvas required. Use $\pi = \frac{22}{7}$.